

INSY7311 A2

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1 FEEDBACK IMPLEMENTATION

The changes applied to Assignment 1 covers:

- the data gathering plan, which was corrected in terms of the methods used for gathering data (instead of other ways of gathering information like sorting and think aloud interviews, in this context, semi-structured interviews and questionnaires was the better approach),
- diving deeper into the introduction of the app,
- redefining usability and user experience,
- and clarifying principles used under the specific design choices chosen.
- Amongst the other changes are the design improvements to the prototypes, by following a more standardised way of displaying a flat design and sketch prototype by using less text and more placeholders.

2 DESIGN PRINCIPLES AND LOW-FIDELITY PROTOTYPE EXPANSION

2.1 Principle 1: Visual Hierarchy

2.1.1 Explanation of Visual Hierarchy

Visual hierarchy is a design strategy used to direct a user's gaze across an interface so that they naturally process elements in order of their true importance (Gordon, 2020). This concept relies heavily on the **principle of scale**, which rules that the dimensions of an element directly communicates its significance and rank within the entire layout (Gordon, 2020). Visual designers tend to encourage the limit of their typography to a maximum of 3 distinct sizes to ensure that the most important information displayed is in the largest format to establish dominance on the screen, and lead the user's eye to the elements of true importance (Gordon, 2020).

2.1.2 Expansion on Design of Low-Fidelity Prototype

Within the "Module Resources" view of the prototype, the layout bypasses the generic entry screens and focuses more on the way that the content is delivered. The module identifier (e.g. "INSY7311") is positioned at the top centre of the screen using the most prominent font size to establish dominance (gives clarification to user on where they navigated to without confusion). Just under that are the secondary medium-sized headers that section based on context like "Past Papers" and "Notes". The individual document names and file details listed below uses the smallest least prominent text size.

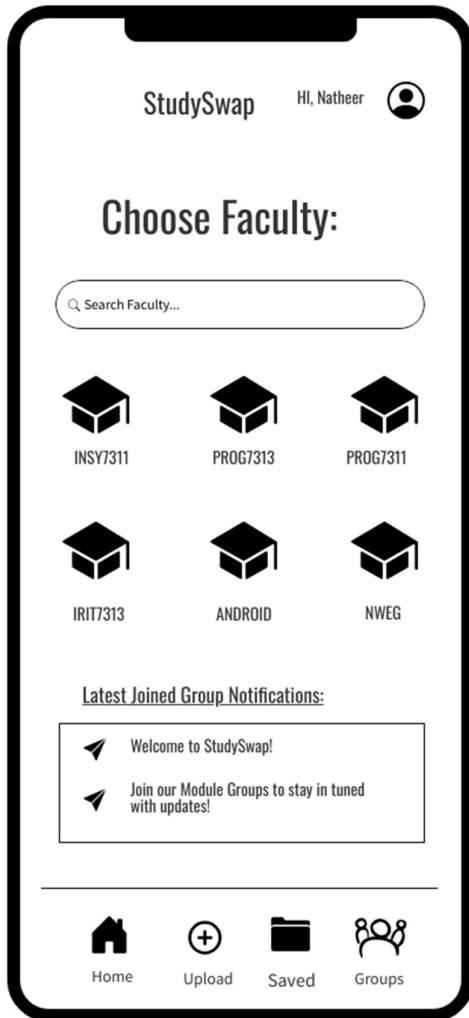


Figure 1 - Home Screen Prototype – Displaying the visual hierarchy of text sizing

2.1.3 Explanation of the Prototype

By using only 3 specific sizes for text, the prototype implements scale to build a strong visual hierarchy (Gordon, 2020). The user's eye instantly gravitates towards the massive module code, giving them immediate confirmation of navigation without confusion (Gordon, 2020). By following the principles of scale and visual hierarchy, the progressively smaller text, in some way it funnels the users vision and attention downward to the contextual headings and then to the file list. This allows the user to process the dense academic information efficiently and in the exact order of priority.

2.2 Principles 2: Contrast

2.2.1 Explanation of Contrast

Contrast is achieved by implementing very distinct visual differences to explicitly signal to a user that these elements serve different structural or interactive purposes (Gordon, 2020). This ensures that the specific content is easily readable and that the actionable elements stand out (Gordon, 2020). A practice that is quite closely related to contrast is the **application of colour (principle relating to contrast)**, but in terms of grayscale shading and light versus dark elements. Proper colour contrast is essential not only for aesthetic purposes to make the visual appealing to the user, but also for maintaining strict usability and accessibility by allowing users to perceive elements against their backgrounds, which in this case, the use of grayscale shading helps easily distinguish between elements (Gordon, 2020).

2.2.2 Expansion on Design of Low-Fidelity Prototype

The “Joined Group Screen” interface gives the users a selection of standard buttons and search bar to search for specific documents, and to select a specific category, and then at the bottom of the screen, there is a primary action button to close the popup notification that is designed with a heavy solid grey fill. This dark grey button is placed directly against a plain background of the app, creating a sharp visual break from the thin outlines of the popup form (Kimball, 2013).

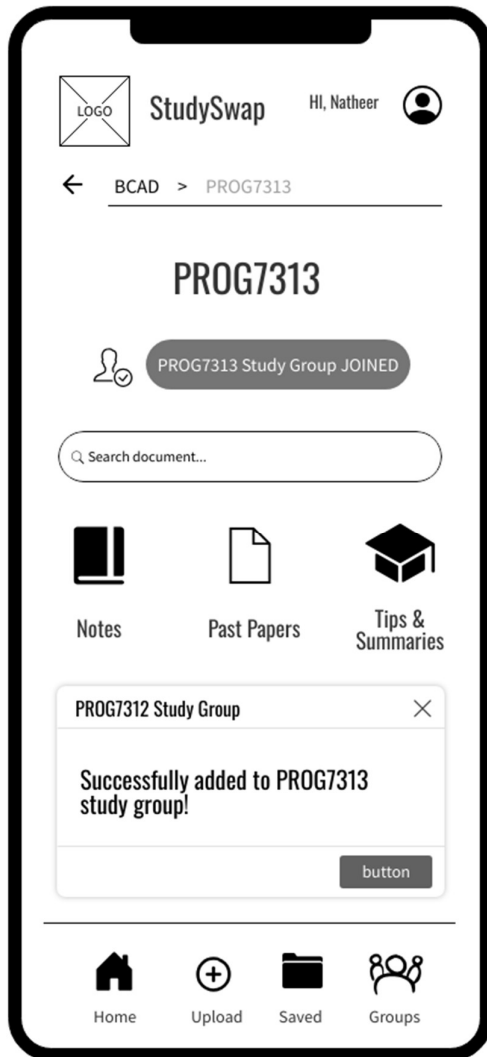


Figure 2 - Joined Group Screen – Displaying Contrast in buttons and background

2.2.3 Explanation of the Prototype

This design choice uses the principle of contrast by manipulating the grayscale tones instead of actual colours. The deliberate juxtaposition of the dark grey buttons against the bright white background clearly tells the user the “Close” button is an interactive action button, distinguishing it functionally from the static background. Even in a low-fidelity state without a full colour palette, making use of the tonal differences (grayscale shading) guarantees that the most critical interaction point is highly visible and is immediately understood by the user. This effectively uses grayscale shading as a structural placeholder to simulate colour contrast that will eventually be implemented in the final design.

2.3 Principle 3: Balance

2.3.1 Explanation of Balance

Balance is accomplished when the visual weight is distributed evenly across an interface, leaving the user with a sense of visual stability (Kimball, 2013). This distribution is often conceptualized around an imaginary central axis, and while it does not dictate mirrored **symmetry (principle relating to balance)** in every instance, symmetrical layouts are a fundamental technique to organize content in a unified way (Kimball, 2013). By establishing symmetrical balance, it ensures that the composition feels structured and harmonious to prevent any single side of the screen from visually overpowering the other (Kimball, 2013).

2.3.2 Expansion on Design of Low-Fidelity Prototype

On the “Upload a Document” screen, the interface follows a linear flow downward but separation is created with the horizontal lines at the top below the navigational path, and at the bottom above the tab-bar-menu. This separation makes it clear to the user the space they should utilize on the screen. The centred alignment of the input fields also gives the user a sense of direction of working top to bottom in terms of what they should be filling in.

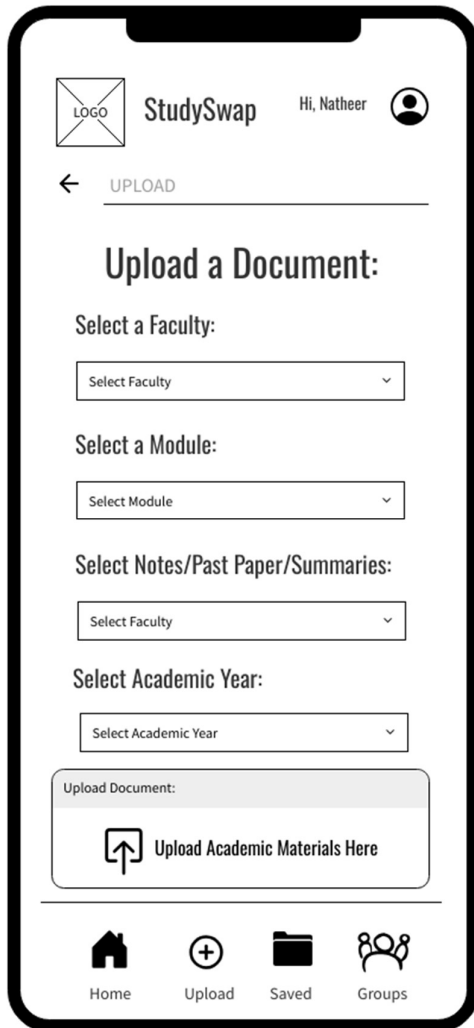


Figure 3 - Upload Document Screen – Showcasing balance and symmetry

2.3.3 Explanation of the Prototype

By strictly centering the alignment of all the input fields and the final upload button, the prototype actively applies the principle of balance to organize the page. If an imaginary vertical axis were drawn directly down the middle of the "Upload a Document" interface, the visual weight of the data-entry boxes would be perfectly and equally distributed on both the left and right sides of this line. This deliberate, symmetrical distribution of the visuals ensures that the composition feels structurally sound and stable.

2.4 Principle 4: Gestalt

2.4.1 Explanation of Gestalt

The **Gestalt framework** describes the psychological tendency of human beings to perceive complex visuals as unified wholes rather than isolated, disconnected parts (Kimball, 2013). A critical component of this phenomenon is the **principle of proximity**, which presumes that items placed physically near each other on a screen are naturally interpreted by the brain as sharing a functional relationship or belonging to the same functional group (Kimball, 2013).

2.4.2 Expansion on Design of Low-Fidelity Prototype

On the apps "Home" screen prototype, the interface shows different pieces of information. At the bottom of the content area, two separate notification alerts are placed physically close together. To further unify them, a solid rectangular bounding box is drawn entirely around these two alerts, clearly separating them from the 3x3 grid of icons positioned above them.

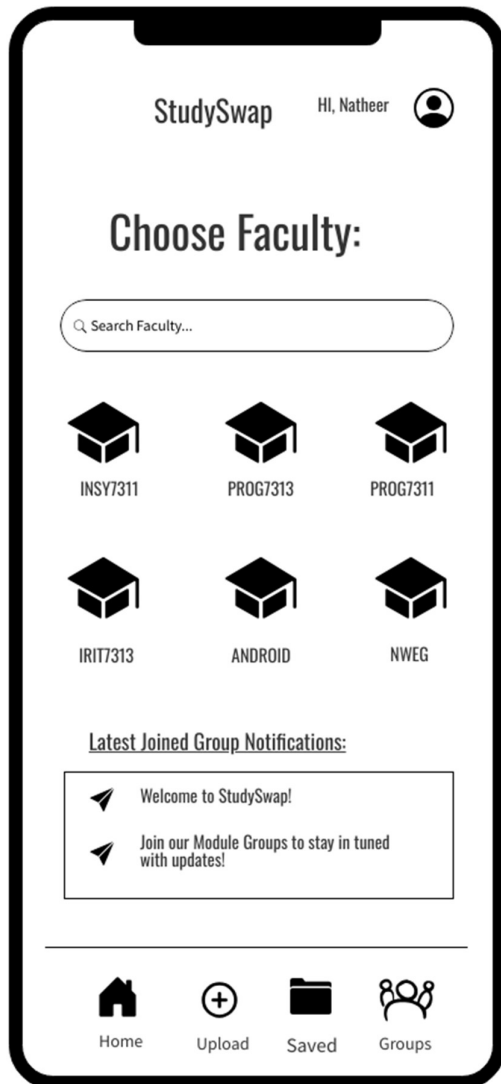


Figure 4 - Notifications Showcasing Gestalt Principle of grouping and Principle of Proximity

2.4.3 Explanation of the Prototype

By packing the notification alerts tightly together and enclosing them within a distinct visual boundary (boxed in), the prototype leverages Gestalt grouping and proximity. The white space separating this box from the faculty icons above forces the user's cognitive system to instantly perceive the notifications as one distinct, related functional block (Kimball, 2013). This grouping drastically reduces the mental strain which enables the user to separate their immediate alerts from the faculty navigation.

2.5 Principle 5: Flat Design

2.5.1 Explanation of Flat Design

Flat design is an interface methodology that emphasizes the extreme minimalism and simplicity in design (Spiliotopoulos, et al., 2018). It makes use of two-dimensional elements, clean lines, and simple shapes, deliberately avoiding skeuomorphic effects like shadows, textures, gradients, or bevels that attempt to mimic a **three-dimensional** physical world (Spiliotopoulos, et al., 2018). So, by removing these intricate details and focusing purely on basic functionality, flat design results in a more streamlined, modern, and intuitive user experience (Spiliotopoulos, et al., 2018).

2.5.2 Expansion on Design of Low-Fidelity Prototype

Across all screens in the prototype (Home, Module, and Upload), a persistent global **navigation tab-bar** is anchored at the bottom. This bar contains icons for "Home," "Upload," "Saved," and "Groups". These interactive icons are drafted strictly as two-dimensional, minimalist line-art. They lack any raised edges, simulated depth, or shaded shadows, laying completely flush against the screen's white background.

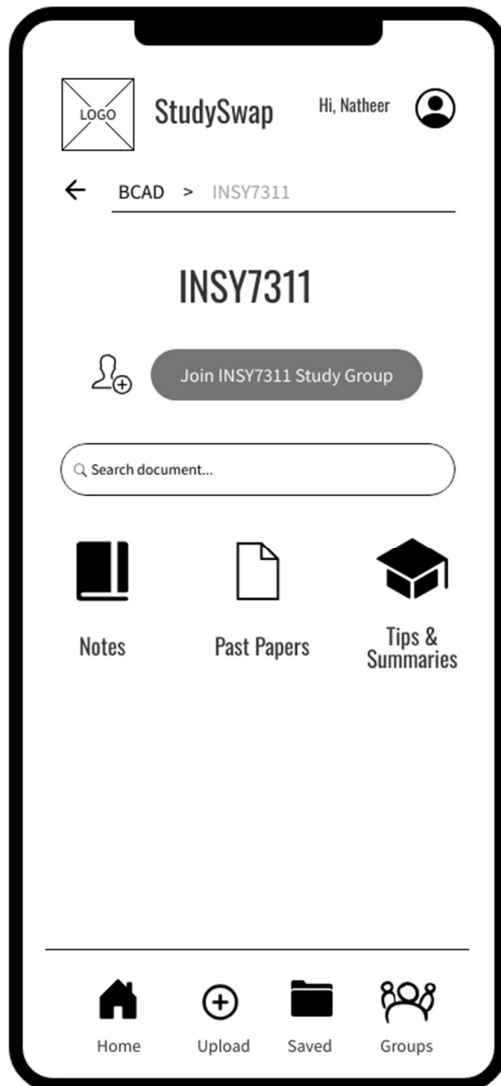


Figure 5 - Flat Designs in Navigation tab-bar

2.5.3 Explanation of the Prototype

By intentionally stripping away three-dimensional effects and mimicking physical button textures, the bottom navigation bar seamlessly embodies the flat design aesthetic. The emphasis on simple, two-dimensional line-mode icons directly highlights the **principle of simplicity**. This minimalistic design removes all the unnecessary visual clutter, ensuring that the user's cognitive attention is focused entirely on the core navigational functionality of the app, allowing for a cleaner and highly efficient mobile interface.

3 EVALUATION PLAN

3.1 Aspect under Evaluation

The evaluation will primarily focus on assessing the general usability and user experience of the low-fidelity prototype's core navigation features and content delivery functions, specifically looking at the hierarchical "Module" screen and the "Upload a Document" interface.

3.2 Evaluation Plan

3.2.1 What

The evaluation will assess the sketches and prototypes of the app to ensure that specific screen functions and the user workflow operate effectively (Rogers, et al., 2023). Ultimately, the goal is to discover whether the hierarchical design actually allows students to do what they need to do, and whether the process of interacting with it is satisfying (Rogers, et al., 2023).

3.2.2 Which

The plan will utilize a mixed approach across two distinct categories of evaluation, which are to conduct studies in controlled settings directly involving participants, and studies conducted in natural settings (Rogers, et al., 2023).

3.2.3 Type

This will be a formative evaluation. Formative evaluations are conducted actively during the ongoing design and development process, rather than at the end, allowing designers to check that the evolving product continues to meet the needs of the people who will use it (Rogers, et al., 2023).

3.2.4 When

This evaluation will take place early in the design lifecycle (Rogers, et al., 2023). Because the app is a new concept, evaluation will occur directly after the initial requirements have been established and embodied into low-fidelity sketches, ensuring that these early interpretations are continuously refined through design-evaluate-redesign cycles (Rogers, et al., 2023).

3.2.5 Where

Testing will take place across a spectrum of settings (Rogers, et al., 2023). To be more specific, it will make use of controlled laboratory environments to carefully measure interactions, as well as natural, real-world settings (in the wild) such as a university campus (Rogers, et al., 2023).

3.2.6 Why

Evaluation is vital to ensure that the final app provides an engaging, pleasing, and simple experience that students will actually want to use (Rogers, et al., 2023). Gathering concrete evaluation data allows the development team to focus on resolving real user problems and to make informed design decisions, rather than the designers simply debating their own personal preferences or waiting until the product is released to discover flaws (Rogers, et al., 2023).

4 EVALUATION METHODS

4.1 Method 1: Usability Testing

4.1.1 Explanation of Usability Testing

Usability testing involves collecting user performance data within a strictly controlled setting such as a usability laboratory (Rogers, et al., 2023). The goal is to determine if the application is usable by having typical users (current university students) perform typical tasks (locating an academic document) (Rogers, et al., 2023). Evaluators measure quantitative performance data, such as the time taken to complete a specific task or the number of errors made while navigating menus, while simultaneously collecting qualitative data by asking the participants to describe what they are doing out loud (Rogers, et al., 2023).

4.1.2 Motivation

This method is highly necessary because it allows the designers to dictate exactly what participants do and when they do it, intentionally stripping away outside real-world distractions like background noise or social interruptions (Rogers, et al., 2023). This level of strict control gives reliable, focused data on fundamental usability problems like whether the centralized alignment on the "Upload a Document" screen actually works, or if the user gets lost navigating the module folders (Rogers, et al., 2023).

4.1.3 How it will be implemented

A sample of typical users (5 to 12 students) will be invited to a quiet, controlled room where they will be given the low-fidelity prototype and be asked to complete predefined tasks, such as to "Upload a past paper for INSY7311" (Rogers, et al., 2023). As they tap through the interface, researchers will utilize the "think-aloud" protocol, asking the students to vocalize their internal thought processes (Rogers, et al., 2023). Evaluators will then record the number of navigational errors made and the time it takes to successfully complete the task, and then follow up with a brief satisfaction questionnaire to gauge their reactions to the design (Rogers, et al., 2023).

4.2 Method 2: In the Wild Studies

4.2.1 Explanation of In The Wild Studies

In the wild studies involve observing how participants use a prototype or technology within their own natural, everyday settings, with little to no direct control or interference from the evaluators (Rogers, et al., 2023). Instead of isolating the user, this method embraces a "messy" environment where activities constantly overlap and are subjected to unpredicted interruptions, allowing researchers to collect data through natural observation and informal interviews (Rogers, et al., 2023).

4.2.2 Motivation

Although controlled usability testing is perfect for finding interface errors, it lacks natural environment validity because a quiet laboratory is an artificial setting (Rogers, et al., 2023). In the wild studies are crucial because they reveal how the app will actually perform in the real world (Rogers, et al., 2023). It allows the researchers to see how students integrate the mobile app into their busy routines, showing its true fitness for an everyday campus context and uncovering unforeseen environmental issues that simply cannot be replicated in a lab (Rogers, et al., 2023).

4.2.3 How it will be implemented

The mobile app prototype will be tested with a group of students to use naturally as they move around their university campus, attend classes, or sit in noisy cafeterias (Rogers, et al., 2023). The researchers will act as nonchalant observers watching how the students interact with the interface when faced with real-world distractions like friends talking or moving between lectures. By following these natural interactions, the researchers will be able to conduct brief, informal interviews to gather information on the student's overall user experience and how effectively the application fits into their real academic environment (Rogers, et al., 2023).

5 REFERENCE LIST

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